

Applicability of the PROSPECT model for estimating protein and cellulose + lignin in fresh leaves

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- Leaf protein and cellulose+lignin (cellig) estimated in fresh leaves at a moderate to good accuracy through inversion with PROSPECT-5.
- Spectral subset [2100,2300] yielded best skill (Figure 1).
- Leaf protein is correlated with nitrogen. Cellig is correlated with leaf carbon. Cellig is important for forest litter decomposition and nutrient cycling.
- Tabulates absorption peaks and mechanisms, with citations, for protein and cellig (Figure 1).

Absorption peaks of protein reported in Curran (1989) and Fourty, Baret, Jacquemoud, Schmuck and Verdebout (1996), and in recalibrated specific absorption coefficients (SAC) from fresh, dry and fresh + dry leaves.

Wavelength (nm) Literature	Wavelength (nm) SAC from fresh leaves	Wavelength (nm) SAC from dry leaves	Wavelength (nm) SAC from fresh + dry leaves	Absorption mechanisms
1020	–	–	–	N–H stretch
1510	1520	1510	1520	N–H stretch, 1st overtone
1730	1750	1730	1730	C–H stretch
1980	–	1980	–	N=H asymmetry
2060	–	2060	–	N=H bend, 2nd overtone, N=H bend, N–H stretch
2130	–	–	–	N–H stretch
2180	2165	2180	2170	N–H bend, 2nd overtone, C–H stretch, C–O stretch, C=O stretch C–N stretch
2240	–	–	–	C–H stretch
2300	2295	2300	2300	N–H bend, C=O stretch, C–H bend, 2nd overtone

Absorption peaks of cellulose + lignin reported in Curran (1989) and Fourty, Baret, Jacquemoud, Schmuck and Verdebout (1996), and in recalibrated specific absorption coefficients (SAC) from fresh, dry and fresh + dry leaves.

Wavelength (nm) Literature	Wavelength (nm) SAC from fresh leaves	Wavelength (nm) SAC from dry leaves	Wavelength (nm) SAC from fresh + dry leaves	Absorption mechanisms
1120	–	–	–	C–H stretch, 2nd overtone
1200	1200	1200	1200	O–H bend, 1st overtone
1420	–	–	–	C–H stretch, C–H deformation
1450	1450	1450	1460	O–H stretch, 1st overtone, C–H stretch, C–H deformation
1490	–	1490	1470	O–H stretch, 1st overtone
1540	–	1540	–	O–H stretch, 1st overtone
1780	1760	–	–	C–H stretch, 1st overtone, O–H stretch, H–O–H deformation
1820	–	–	–	O–H stretch, C–O stretch, 2nd overtone
2100	2100	2100	2100	O=H bend, C–O stretch, C–O–C stretch, 3rd overtone
2270	–	–	–	C–H stretch, O–H stretch, CH ₂ bend, CH ₂ stretch
2280	–	–	–	C–H stretch, CH ₂ deformation
2340	2340	2340	2340	C–H stretch, O–H deformation, C–H deformation, O–H stretch

Selected spectral subsets for model inversion to estimate leaf protein and cellulose + lignin content based on absorption features from the literature, global sensitivity analysis and stepwise multiple linear regressions.

Spectral subset	Wavelengths (nm)
1	1100–2500
2	1600–1800, 2100–2300
3	2100–2300
4	1020, 1510, 1730, 1980, 2060, 2130, 2180, 2240, 2300
5	1120, 1200, 1420, 1450, 1490, 1540, 1780, 1820, 2100, 2270, 2280, 2340
6 ^a	1345, 2120, 2295, 2300
6 ^b	1320, 1450, 2015, 2290
7 ^a	1150, 1185, 1220, 1295
7 ^b	2000, 2150, 2170, 2255

Spectral subset 2 and 3 are selected based on global sensitivity analysis. 4 and 5 are associated with protein absorption features and cellulose + lignin absorption features, respectively. See (Curran, 1989; Fourty, Baret, Jacquemoud, Schmuck & Verdebout, 1996). 6^a and 7^a are selected spectral subsets for protein and cellulose + lignin, respectively, based on stepwise multiple linear regressions on fresh leaves. 6^b and 7^b are selected spectral subsets for protein and cellulose + lignin, respectively, based on stepwise multiple linear regressions on dry leaves.

Figure 1: Literature absorption peaks for protein (top) and cellulose + lignin (middle), and the spectral subsets selected for model inversion from them (bottom).